Ryan Maxin

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EDUCATION

University of Waterloo **Sep 2021 – Sep 2026**
Bachelor of Computer Science, Co-op *Waterloo, Ontario*

- Graduated on the **Dean's Honours List** with a cumulative average of 87.1%

EXPERIENCE

NVIDIA **Oct 2026 – Present**
GPU Compiler Software Engineer *Austin, Texas*

NVIDIA **Sep 2025 – Dec 2025**
GPU Compiler Engineer Intern *Austin, Texas*

- Unified **PTX** load-constant instruction representation in the **CUDA** compiler on **Hopper+ GPUs** by consolidating dual `MOV`/`READ` forms into a single canonical representation, enabling consistent downstream analysis and optimization across the backend.
- Implemented the unification in **C++** via a full backend audit and empirical validation, recovering baseline performance and delivering **5%+** improvements across dozens of production GPU kernels while surfacing latent correctness issues.

Qualcomm **May 2025 – Aug 2025**
ML Compiler Engineer Intern *Austin, Texas*

- Architected a sharding algorithm for transformer-based **LLMs** in **PyTorch**, partitioning MLP and attention blocks across a fixed number of **NSP cores** to enhance parallelism and enable scalable on-device inference.
- Lowered the sharded graph into the NSP compiler, modifying its core-mapping logic in **C++** to apply mappings produced by the partitioning algorithm, achieving up to a **50% reduction** in inter-core communication overhead.
- Engineered custom **all-reduce/all-gather** primitives at the **PyTorch** level to balance synchronization across compute units, **maximizing** core utilization.

AMD **Jan 2025 – Apr 2025**
Software Engineer Intern *Toronto, Ontario*

- Implemented **Vulkan**, **DirectX**, **OpenGL**, and **OpenCL** compute kernels to enable cross-platform color-space and scaling conversions for **YUV 4:4:4** and **4:2:2** formats within AMD's Advanced Media Framework.
- Unified and optimized host-side conversion paths in **C++**, eliminating **~1,500** lines of redundant code and precomputing shader parameters to improve runtime performance by **~6%**.
- Built an automated testing pipeline validating **20+** format conversions across **tens of thousands** of configuration combinations, performing statistical **SSIM/PSNR** analysis to ensure high-fidelity and consistent results.

Huawei **May 2024 – Aug 2024**
Compiler Engineer Intern *Toronto, Ontario*

- Enhanced loop tiling compiler optimization in **LLVM** using **C++**, successfully incorporating it as a default pass and boosting runtime performance of high-performance computing workloads by up to **7%**.
- Implemented an **LLVM** compiler pass in **C++** to optimize high-bandwidth memory allocation in **multithreaded** programs by leveraging **IR** pointer analysis, reducing memory contention by as much as **60%**.
- Integrated **profile-guided optimization** into the pass using cache miss data from Linux **perf** to further improve program throughput.

WSIB Innovation Lab **Sep 2023 – Dec 2023**
Full Stack Engineer Intern *Waterloo, Ontario*

- Architected a **generative AI** tool to enable conversational interaction with **SQL** databases.
- Leveraged the unique strengths of different quantized and fine-tuned **LLMs** using **Python** and **LangChain** to achieve parity with **GPT-4** in internal database QA, effectively addressing OpenAI data privacy concerns.
- Deployed the AI to **Azure VMs** and presented the results to a **150-person audience** in the CBIA branch.

Infinera **Sep 2022 – Dec 2022**
Firmware Engineer Intern *Ottawa, Ontario*

- Created a CLI API in **C++** using **eRPC** to enable low-level runtime reconfiguration of digital subcarriers, eliminating a **10-minute** recompile time for developers.
- Utilized the Atlassian API, **Pandas**, and **Matplotlib** in **Python** to display real-time graphical views of production build data from **Jenkins** CI on a Confluence page, allowing the team to identify and fix resource inefficiencies on hardware limited to 32MB of memory.

University of Waterloo **Jan 2022 – Apr 2022**
Course & Technical Support Assistant *Waterloo, Ontario*

- Engineered a **JavaScript** algorithm to process **thousands** of raw Sentinel-1 Arctic satellite radar images in Google Earth Engine to determine melt/freeze onset dates for lakes and rivers in Canada.

RESEARCH

University of Waterloo **Apr 2026 – Aug 2026**
Undergraduate Research Assistant *Waterloo, Ontario*

- Developed a prototype redesign of the **Scala 3 compiler**'s global initialization checker in **Scala**, implementing a primarily flow-insensitive **abstract interpretation** over global-object dependencies, reachable definitions, and initialization cycles using fixed-point computation.
- Implemented **interprocedural analysis** across dynamic dispatch, lazy values, closures, mutable state, and class initialization.

PROJECTS

Joos Compiler

- Built an end-to-end **Java compiler** in **OCaml**, implementing an **LALR(1) parser**, AST construction, name resolution, type checking, and static analysis for multi-file programs.
- Generated **32-bit x86 assembly** with object layouts, virtual/interface dispatch, runtime subtype checking, arrays, and a stack-based calling convention.

Black Hole Renderer

- Built a real-time **WebGL 2/GLSL** renderer that ray traces light through **Schwarzschild spacetime**, numerically integrating geodesics with **RK4** to render gravitational lensing and an accretion disk.
- Optimized GPU ray tracing with adaptive integration and early termination, achieving up to an **8.9x speedup** and **4.68x fewer integration steps** while maintaining sub-pixel accuracy.

Sorting Algorithm Visualizer

- Developed an interactive visualizer of **6** popular sorting algorithms using **React** and **Material UI**.

Realm Tunes

- Created a feature-rich Discord music bot using **Python** and the discord.py API.
- Deployed Realm Tunes to an **Ubuntu** server, providing on-demand music to **350 users** across **7 servers**.

SKILLS

Languages: C/C++, Python, Scala, OCaml, Java, JavaScript, TypeScript, Kotlin, HTML/CSS, SQL, x86 Assembly, VHDL
Compilers & GPU: LLVM, CUDA, PTX, Vulkan, DirectX, OpenGL, OpenCL, WebGL, GLSL
Tools & Frameworks: Git, Perforce, Linux, PyTorch, React, Docker, perf, Jenkins, Azure, Firebase, Jira